

Name: Grant Harvey

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Honor Pledge: I pledge that the only resources I used to complete this homework were the course textbook, the course slides, or my personal course notes. I did not confer with any other student, consult any solutions manuals, printed or on-line, nor did I do any Internet searches to solve any problems for this homework assignment. The solutions below are my own work.

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Cyber Security Fundamentals – CHAPTER 1 HOMEWORK

70 points total

- 1.4** [25] For each of the following assets, assign a low, moderate, or high impact level for the loss of confidentiality, availability, and integrity, respectively. Confidentiality, availability, and integrity are each to be assigned an impact level. **Justify your answers.**
- a.** An organization managing public information on its Web server.
 - b.** A law enforcement organization managing extremely sensitive investigative information.
 - c.** A financial organization managing routine administrative information (not privacy related information).
 - d.** An information system used for large acquisitions in a contracting organization contains both sensitive, pre-solicitation phase contract information and routine administrative information. Assess the impact for the two data sets separately and the information system as a whole.
 - e.** A power plant contains a SCADA (supervisory control and data acquisition) system controlling the distribution of electric power for a large military installation. The SCADA system contains both real-time sensor data and routine administrative information. Assess the impact for the two data sets separately and the information system as a whole.

Solution:

	<u>Confidentiality</u>	<u>Integrity</u>	<u>Availability</u>
A: An organization managing public information on its Web server.	Low since data is public anyway so it shouldn't matter if people see everything.	Moderate because if someone were to get into the system and change the public information to be false it could be an issue.	Moderate because if the website goes down then users can't access the information.
b. A law enforcement organization managing extremely sensitive investigative information.	High because its sensitive information that we want to keep safe so we don't want it leaked to the world	High because if someone can get into this organization and change the information it would be terrible. Like if they	Moderate because while investigators need to access this info, a temporary outage wouldn't destroy their work.

		compromised investigations.	
c. A financial organization managing routine administrative information (not privacy related information).	Low because its not sensitive information so a leak wouldn't be the worst possible outcome.	Moderate because modification could cause some incorrect records, but wouldn't destroy any operations.	Moderate because an outage would delay any records, but wouldn't cause any permanent damage.
d: pre-solicitation phase contract information	High because it could compromise the acquisition process	High because they could alter the acquisition requirements, pricing information, or other sensitive info	Moderate because it could delay massive acquisitions but wouldn't overly damage anything
d: routine administrative information	Low because routine administrative info usually wouldn't be highly sensitive	Moderate because altered info could disrupt and slow down operations but wouldn't destroy anything.	Moderate because it could delay the routine but wouldn't cause any long lasting issues.
d: information system as a whole.	High because it would include all the sensitive information	High because if you could change or remove any high sensitivity information that would cause a lot of issues.	Moderate because while the system needs to be accessed a lot, a temporary down time wouldn't cause major damage.
E: real-time sensor data	High because a leak of operational information could reveal things about the installation's power infrastructure which in turn foreign powers could use.	High because unauthorized modification could cause incorrect control actions or disrupt power distribution	High because loss of availability could prevent operators from monitoring or controlling the power system and potentially cause big safety issues.
e: routine administrative information	Low because routine administrative info usually wouldn't be highly sensitive	Moderate because altered info could disrupt and slow down operations but wouldn't destroy anything.	Moderate because it could delay the routine but wouldn't cause any long lasting issues.
e: information system as a whole.	High because it would include all the sensitive information	High because if you could change or remove any high sensitivity information that would cause a lot of issues. Also it could directly affect the operation of critical infrastructure.	High Because imagine you have an outage and can't install highly necessary military tools. But also it could be a short outage at a time when its not an issue, so it really just depends on when it happens.

1.5 [20] Basic Concepts Knowledge Checking.

- a) [10 points] What are the Vulnerabilities, Threats, and Attacks in information security? How can we categorize the Attacks?
- b) [7 points] What is the security policy? What is the security mechanism? What are three categories of security mechanisms used to protect systems?
- c) [3 points] List and briefly describe the three categories of attack surface.

Solution:

a:

Vulnerability = A weakness or flaw with the system

Threat = The potential action you could take to exploit a vulnerability

Attack = acting out a threat

We can categorize attacks as Passive, Active, Insider, or Outsider.

Passive = gaining info but not harming the system

Active = Alter system to somehow affect the operation

Insider = initiated the attack inside the security

Outsider = initiated the attack outside the security

b:

Security Policy says what isn't and what is allowed. A Security Mechanism is a system put in place to stop threats like 2-factor authentication or a firewall.

Three Security Mechanism Categories:

1. Prevention = stop attackers from violating a security mechanism
2. Detection = see every security attack
3. Recovery = stop attack, assess, and repair; or continue to function even if attack succeeds

c:

Network Attack Surface = Vulnerabilities over a wide-area network or the internet

Software Attack Surface = Vulnerabilities in application, utility, or operation system code

Human Attack Surface = Vulnerabilities created by Users such as human error, physical attacks, or social engineering like phishing emails.

1.6. [25] Develop an attack tree for gaining access to the contents of a physical safe.

Solution:

